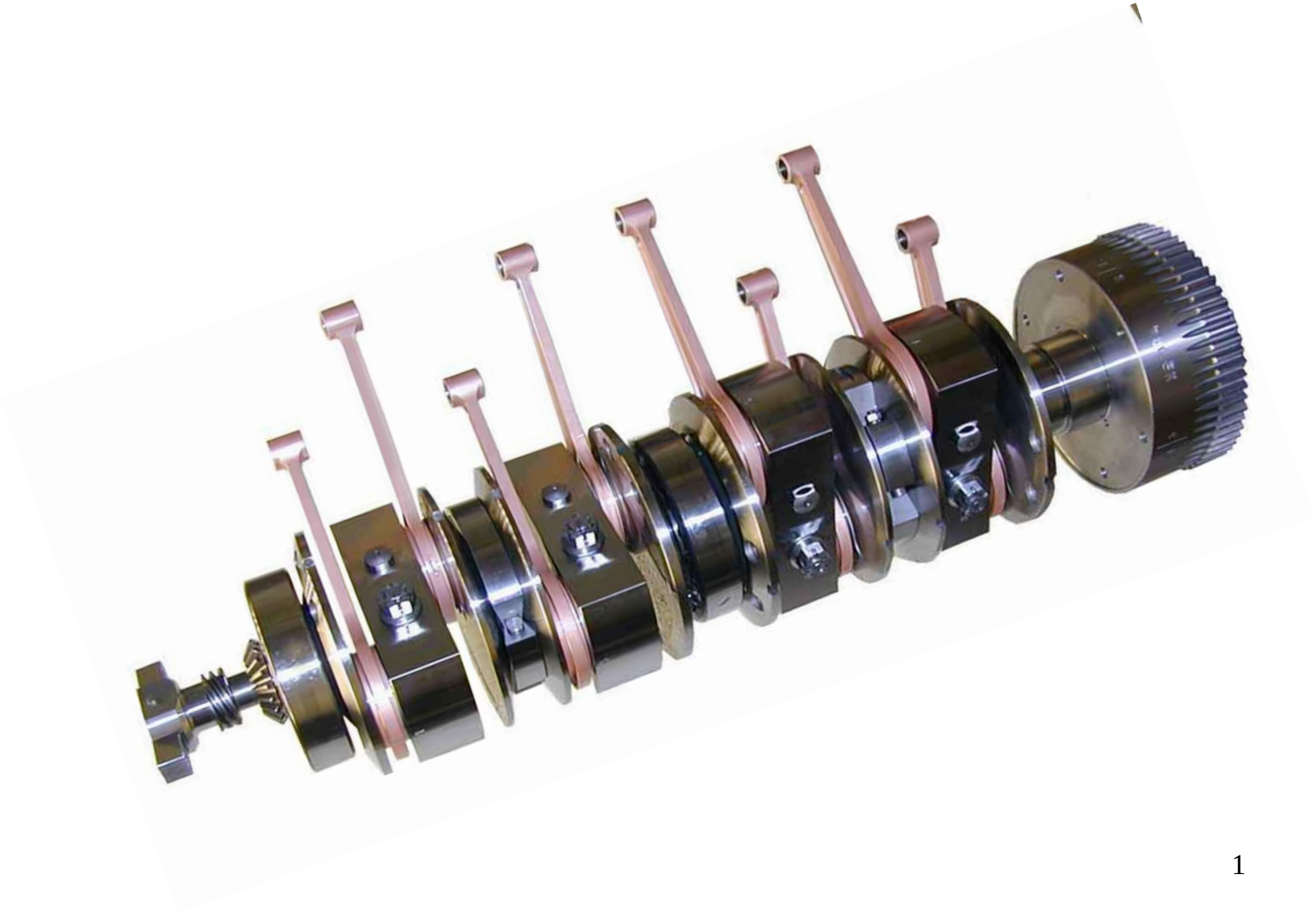
Limits, Fits and Tolerance



SYSTEM OF FITS AND TOLERANCES

- **The standard reference temperature** for industrial length measurements is 20 ^{o}C (ISO 1:1975), and, consequently, for dimensions defined by the system
- **Limits:** The 2 **extreme permissible sizes of a part** between which the actual size is contained
- Tolerance: The **total amount a specific dimension is permitted to vary**
- So, **tolerance** is the difference between maximum limit and minimum limit of size
- **Fit**: the relationship existing between two parts which are to be assembled with respect to the difference on their sizes before assembly
- Fits are generally described as **"shaft and hole"**
- ISO is the internationally accepted standard for defining **fits but ANSI** is often still used in America

Limits & Tolerance
Example 1

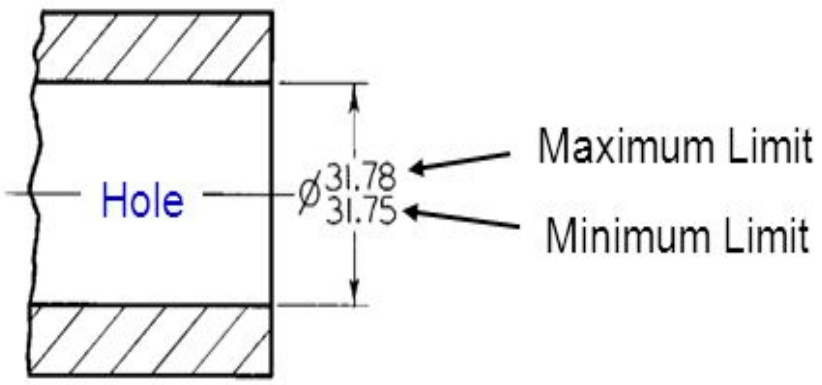
- **Limits: maximum and minimum dimensions for a given size**

  Example, 20 ± 0.1.

- The upper and lower limits of size are 20.1 mm and 19.9 mm respectively.
- **Tolerance: algebraic difference between the upper** and lower limit of size
- In this case, Tolerance = 20.1 – 19.9

  = 0.2 mm

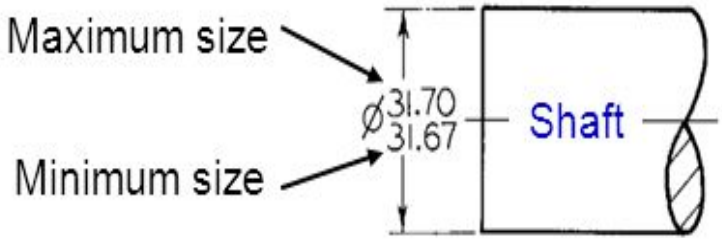
Limits & Tolerance
Example 2



Tolerance = total amount of variation

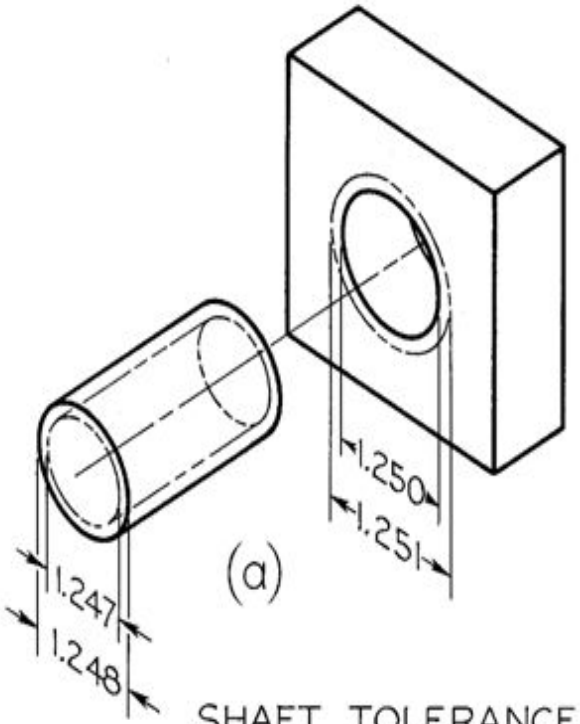
31.78 – 31.75 = 0.03

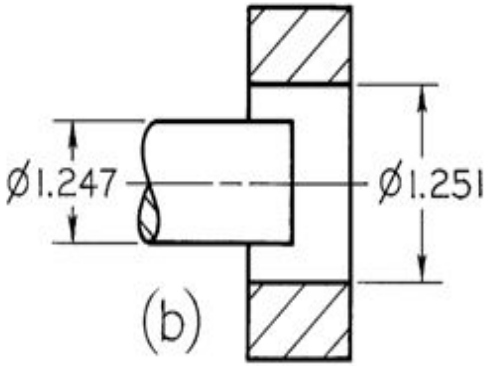
Dimension could be 31.78 or 31.75
or any value between them



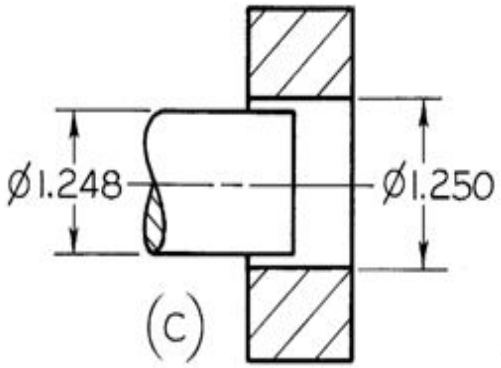
Tolerance = 31.70 – 31.67 = 0.03

Fits
Example 3





LOOSEST FIT
Smallest Shaft in Largest Hole



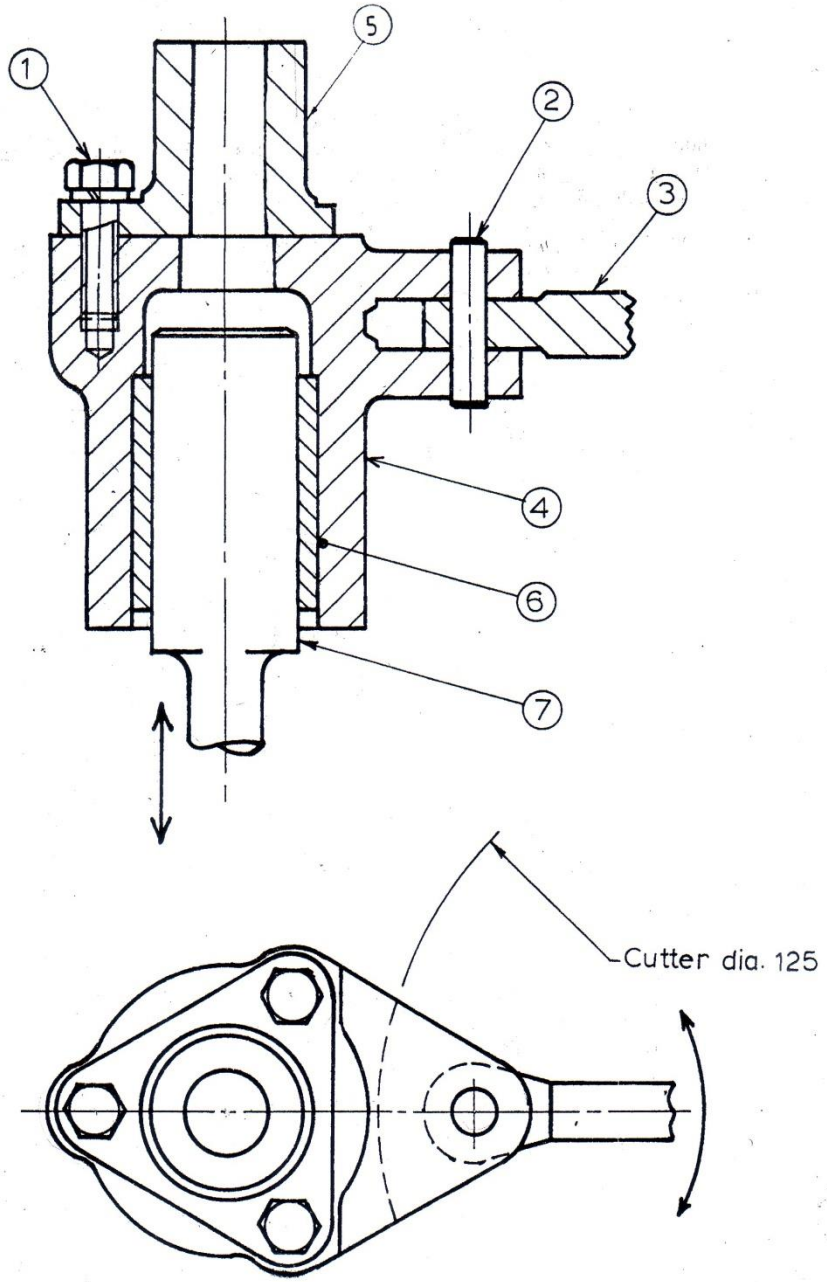
TIGHTEST FIT
Largest Shaft in Smallest Hole

SHAFT TOLERANCE = 1.248 − 1.247 = .001
HOLE TOLERANCE = 1.251 − 1.250 = .001

ALLOWANCE = 1.250 − 1.248 = .002
MAX CLEARANCE = 1.251 − 1.247 = .004

Fits and limits: Points to note

- In practice, no component can be manufactured to an exact size
- To ensure that an assembly will **function correctly**, its component parts must fit together in a predictable manner.



TOLERANCE

Classes of Tolerance

There are 3 main classes:

1. **Tolerance for size (dimensional tolerance)**
2. **Tolerance for position (location)**
3. **Tolerance for form**

- There is also a **supplementary classification that define shape related parameters** such as **flatness, straightness, parallelism, squareness or angularity**, etc.
- This is called Geometric tolerance

Dimensional Tolerance:

- Manufacturing costs increases considerably as tolerances zones decrease.
- Hence, general tolerances are often adopted, which allow reasonable limits for every day applications
- Hence, maximum possible tolerances must be recommended wherever possible

- General tolerances:

Manufacturing process	General tolerances (mm)
Drilling	+ 0.125
Turning – diameters	± 0.15
Grinding – surfaces	± 0.025

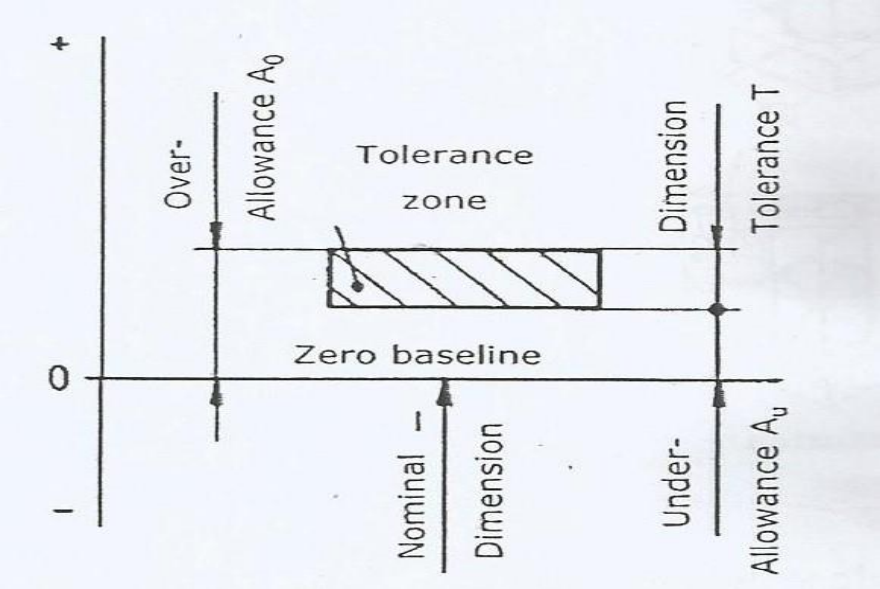
Dimensional (size) Tolerance, Limits and fits

- Manufacturing limits are specified relative to the basic size and expressed unilaterally or bilaterally according to the system of limits adopted.

(i)Nominal size: the dimension to which a part is to be made if it were possible to manufacture to exact size., i.e., say ϕ35 or ϕ60

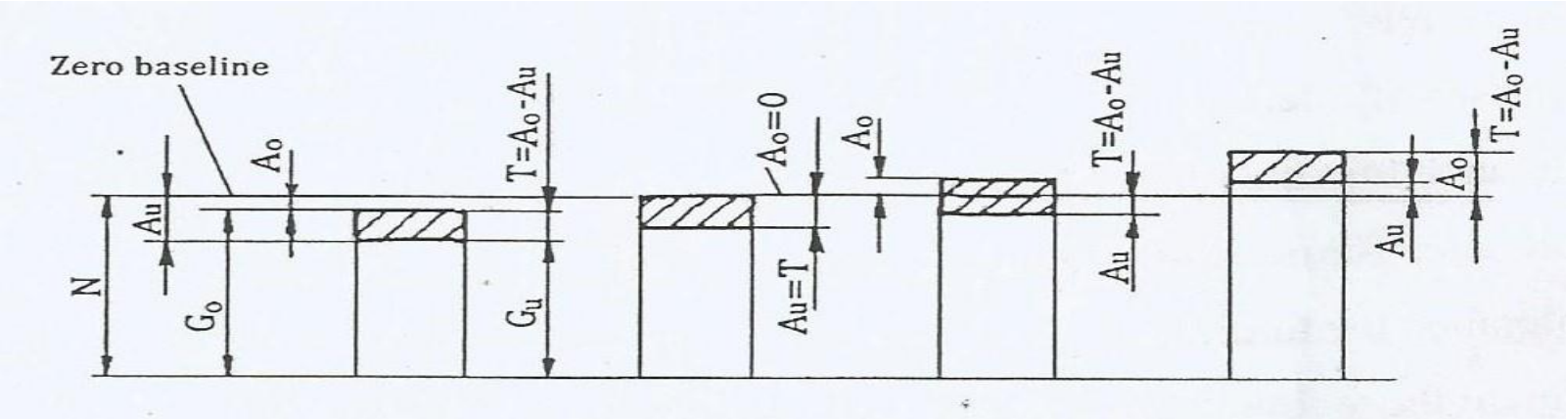
(ii)Basic size: the nominal size to which deviations that determine the fit are applied, written as 35.000 (i.e. when the nominal size is converted to decimals)

(iii)For mating parts, the basic size **is equal.**



Dimension (size) Tolerance, Limits and fits

iv) Deviation: the algebraic amount by which a size is greater (+) or less (-) than the basic size



Upper deviation: the **difference between the maximum limit of size and the corresponding basic size**

Designation ES for HOLE and es for SHAFT

Lower deviation: the **difference between the minimum limit of size and the corresponding basic size**

Designation EI for HOLE and ei for SHAFT

Limit of size: the minimum (lower) and maximum (upper) sizes for a dimension that is permitted

$$\left.\begin{matrix} 25.010 \ \ \max.\ size \\ 24.000 \ \ \min.\ size \end{matrix}\right\} Tolerance\ field$$

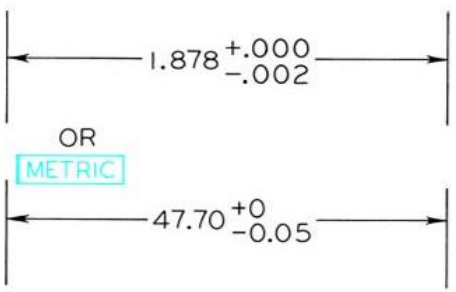
Bilateral Tolerance:
allows variations in both directions from **the basic size**, e.g. 25±0.010

Unilateral Tolerance: Allows variations in only one direction from the **basic size**. **These tolerance lie wholly above the basic size** for a HOLE dimensioning and below the design size for SHAFT dimensioning

$$\begin{Bmatrix} 90+0.02 \\ 90+0.00 \end{Bmatrix} for\ Holes$$

$$\begin{Bmatrix} 90-0.020 \\ 90-0.150 \end{Bmatrix} for\ Shaft$$

Unilateral tolerances



Free Dimensions:

- Free dimensions are dimensions that are not involved in mating of parts
- They are applied to machine parts that are produced by means of either metallic removal or forming
- They may involve lengths, curvature (radius, diameter) and angles
- They are divided into four classes

$$\begin{Bmatrix} f-fine \\ m-medium \\ c-coarse \\ vc-very\ close \end{Bmatrix}$$

- The deviations are functions of the nominal size range of the length, curvature or angle
- Is **normally expressed as General Tolerance and it is written within Title Block**
- Eg.: ALL METRIC DIMENSIONS TO BE HELD TO ±0.05

This **implies dimension 65 will have upper limit 65.05 and lower limit 64.95**

Free Dimensions

DIN7168 is a general tolerance standard for linear and angular dimensions.

Table 1. Permissible deviations for linear dimensions except for external radii and chamfer heights

Tolerance class	Permissible deviations, in mm, for nominal sizes, in mm,											
	from 0.5 to 3	over 3 up to 6	over 6 up to 30	over 30 up to 120	over 120 up to 400	over 400 up to 1000	over 1000 up to 2000	over 2000 up to 4000	over 4000 up to 8000	over 8000 up to 12000	over 12000 up to 16000	over 16000 up to 20000
f (fine)	±0.05	±0.05	±0.1	±0.15	±0.2	±0.3	±0.5	±0.8	—	—	—	
m (medium)	±0.1	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	±4	±5	±6
g (coarse)	±0.15	±0.2	±0.5	±0.8	±1.2	±2	±3	±4	±5	±6	±7	±8
sg (very coarse)	—	±0.5	±1	±1.5	±2	±3	±4	±6	±8	±10	±12	±12

For nominal sizes below 0.5 mm, the deviations shall be indicated adjacent to the relevant nominal size.

Free Dimensions

Table 2. Permissible deviations for external radii and chamfer heights

Tolerance class	Permissible deviations, in mm, for nominal sizes, in mm,				
	from 0.5 to 3	over 3 up to 6	over 6 up to 30	over 30 up to 120	over 120 up to 400
F (fine)	±0.2	±0.5	±1	±2	±4
M (medium)					
G (coarse)	±0.2	±1	±2	±4	±8
Sg (very coarse)					

Free Dimensions

Table 3. Permissible deviations for angular dimensions

Tolerance class	Permissible deviations, in units of angle, for nominal sizes of the shorter leg, in mm.				
	over 10	over 10 up to 50	over 50 up to 120	over 120 up to 400	over 400
F (fine)	±1°	±30′	±20′	±10′	±5′
M (medium)					
G (coarse)	±1°30′	±50′	±25′	±15′	±10′
Sg (very coarse)	±3°	±2°	±1°	±30′	±20′

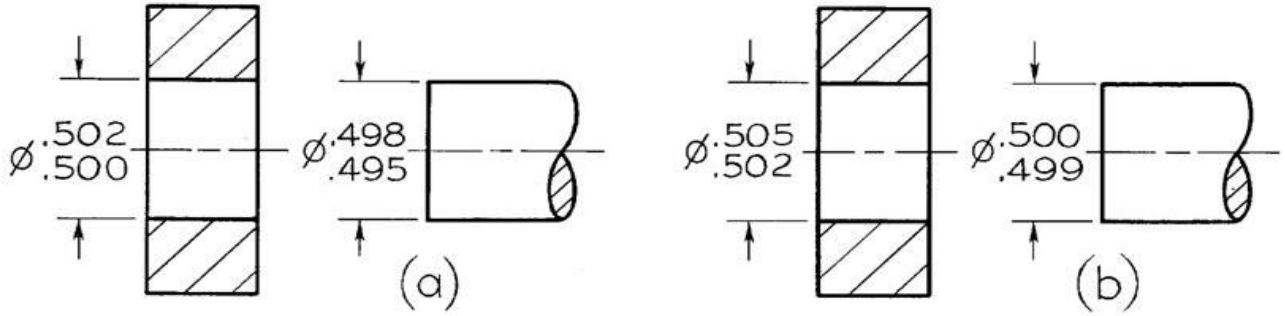
MATING DIMENSIONS:

Basic hole and shaft systems

Systems of Fits:

i) **Basic Hole System:** **Minimum hole size is taken** as the **basic size** and the variations are permitted on the shaft to obtain the desired fit (normal use in Engineering Works)

i) **Basic Shaft System**: **The maximum shaft size is taken** as the **basic size** and the hole sizes are varied to obtain the desired fit (special for industries)

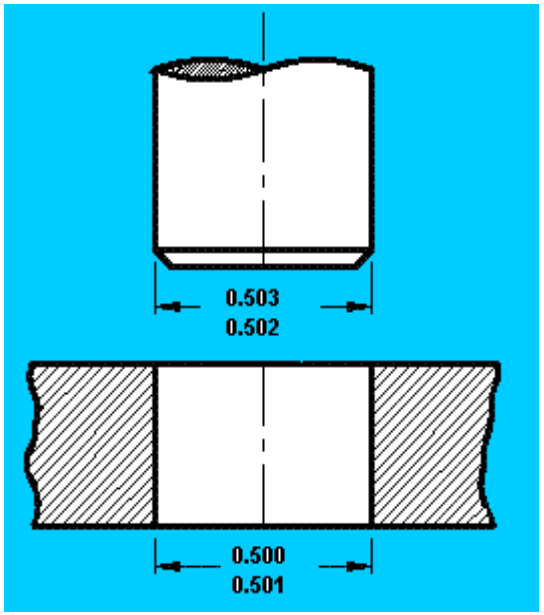


BASIC HOLE FIT

BASIC SHAFT FIT

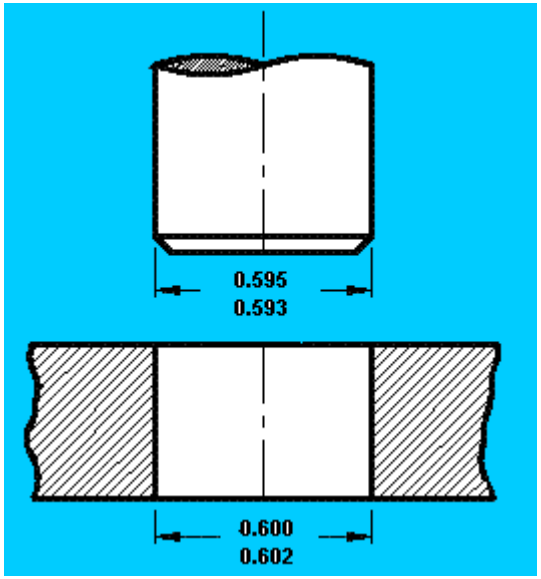
Fits

Allowance: is the minimum clearance space (or maximum interference) intended between mating parts



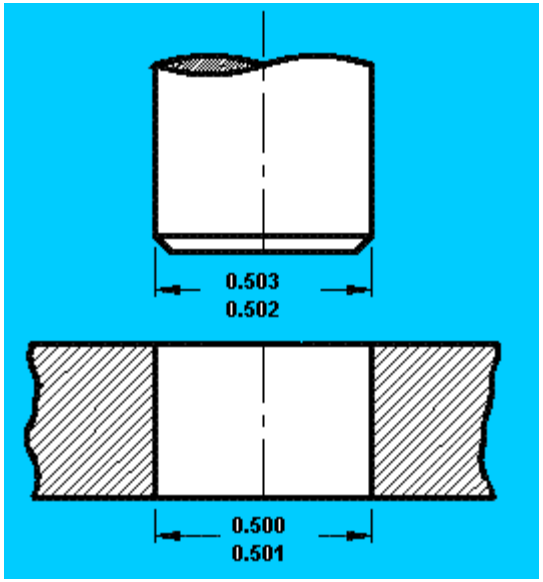
Description of FITS: three main types of fits:

1) Clearance fit: when the shaft is SMALLER than the hole size
2) Interference fit: when the shaft is BIGGER than the hole size
3) Transitional fit: when both shaft and hole sizes are about equal.
This occurs between Clearance and Interference fits



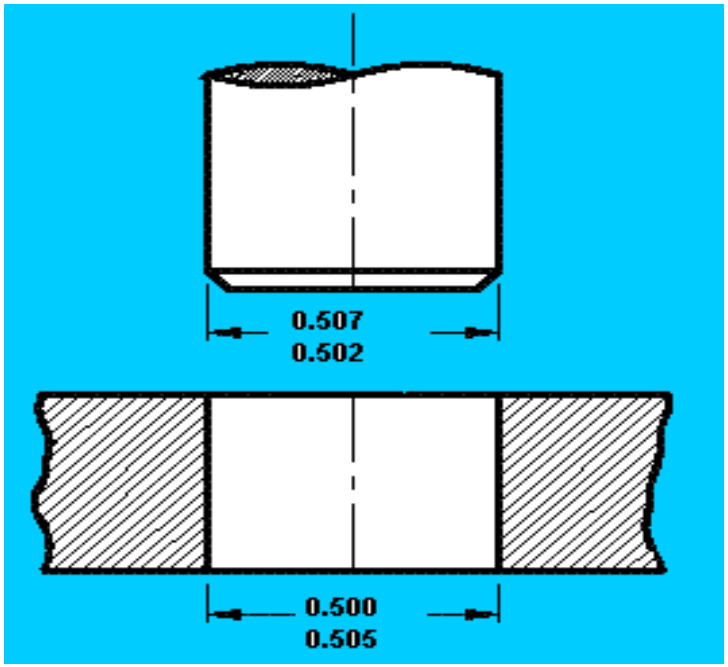
A-CLEARANCE FIT : A **type of fit in which one part fits easily into another with a resulting clearance**. Example,

A clearance fit. The shaft is always smaller than the hole
Tolerance on shaft : 0.002
Tolerance on hole : 0.002
minimum clearance : 0.600 - 0.595 = 0.005 inch
maximum clearance : 0.602 - 0.593= 0.009 inch
The 0.005 clearance is for the tightest possible fit.



INTERFERENCE FIT : A type of fit in which one part must be forcibly fitted into another. Example:

when the shaft is always larger in diameter than the hole, parts must be assembled by pressure or heat expansion.
Tolerance on shaft : 0.001
Tolerance on hole : 0.001
minimum clearance : 0.500 - 0.503= -0.003 in (the **tightest fit 0.003 in interference)**
maximum clearance : 0.501 - 0.502 = -0.001 in (the **loosest fit 0.001 in interference)**
Maximum clearance=Minimum interference
Minimum clearance=Maximum interference



TRANSITION FIT : A type of fit in which **loosest case provides a clearance fit** and the **tightest case gives an interference fit.** Example:

A **transition fit exist when the maximum clearance is positive and the minimum clearance is negative**

Tolerance on shaft : 0.005

Tolerance on hole : 0.005

minimum clearance : 0.500 - 0.507 = -0.007 inch

The **tightest fit is 0.007 in interference.**

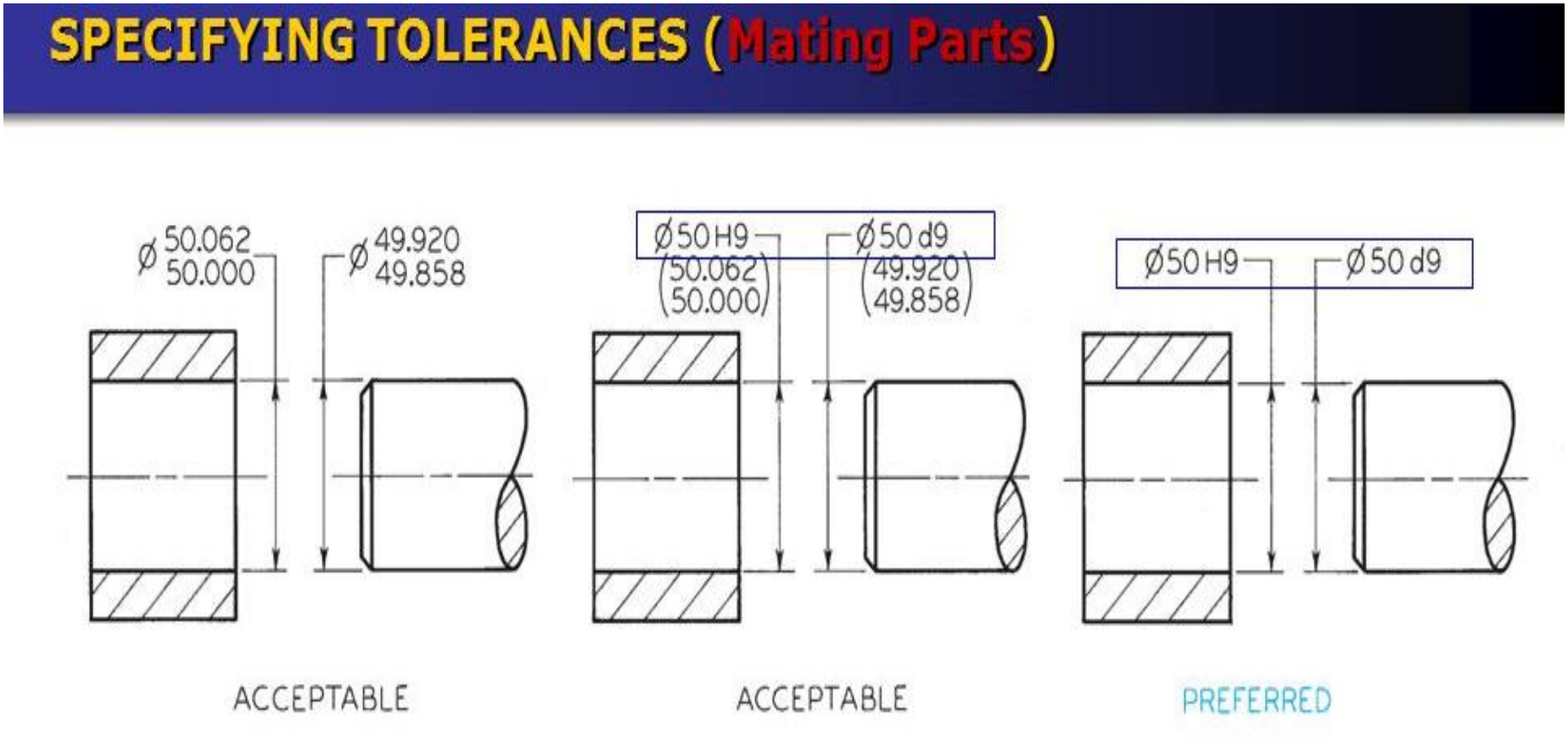
maximum clearance : 0.505 - 0.502 = 0.003 inch

The loosest fit is 0.003 in clearance

MATING DIMENSIONS:

Specification of FITS:

- In the standards, letters (positions) and numbers (size of the tolerance filed) are used to describe combination of some particular deviation and tolerance



SPECIFYING TOLERANCE (Mating Parts)

- METRIC SYSTEM OF TOLERANCES AND FITS (ISO SYMBOLS)

Hole base system:

- Hole: H (uppercase)
- Shaft: (c, d, f, g, h), (k, n), (p, s, u) (lowercase)

Shaft base system:

- Hole: h (lowercase)
- Shaft: (C, D, F, G, H), (K, N), (P, S, U) (uppercase)
- **Each letter has a meaning according to the class of fit.**

MATING DIMENSIONS:

CLASSES OF FITS (ISO SYMBOLS)

Clearance fits:

1) **Loose running fit: H11/c11**

   Wide commercial tolerances. Precision is not important.

2) **Free running: H9/d9**

   Accuracy is not important, or large temp. variation, or both conditions.

3) **Close running: H8/f7**

   Running on accurate machines, accurate locations at moderate speeds and journal pressure. Accurate location and minimum play are desired.

4) **Sliding fit: H7/g6**

   For easy move and turn (no free motion), and accurate location.

5) **Locational fit (precision clearance fit): H7/h6**

   Location of stationary parts (static applications).

MATING DIMENSIONS:

CLASSES OF FITS (Clearance fits)

Dimensions are in millimeters.

Basic Size		Loose Running			Free Running			Close Running			Sliding			Locational Clearance		
		Hole H11	Shaft c11	Fit	Hole H9	Shaft d9	Fit	Hole H8	f7	Fit	Hole H7	Shaft g6	Fit	Hole H7	Shaft h6	Fit
1	Max	1.060	0.940	0.180	1.025	0.980	0.070	1.014	0.994	0.030	1.010	0.998	0.018	1.010	1.000	0.016
	Min	1.060	0.880	0.060	1.000	0.955	0.020	1.000	0.984	0.006	1.000	0.992	0.002	1.000	0.994	0.000
1.2	Max	1.260	1.140	0.180	1.225	1.180	0.070	1.214	1.194	0.030	1.210	1.198	0.018	1.210	1.200	0.016
	Min	1.200	1.080	0.060	1.200	1.155	0.020	1.200	1.184	0.036	1.200	1.192	0.002	1.200	1.194	0.000
1.6	Max	1.660	1.540	0.180	1.625	1.580	0.070	1.614	1.594	0.030	1.610	1.598	0.018	1.610	1.600	0.016
	Min	1.600	1.480	0.060	1.600	1.555	0.020	1.600	1.584	0.006	1.600	1.592	0.002	1.600	1.594	0.000
2	Max	2.060	1.940	0.180	2.025	1.980	0.070	2.014	1.994	0.030	2.010	1.998	0.018	2.010	2.000	0.016
	Min	2.000	1.880	0.060	2.000	1.955	0.020	2.000	1.984	0.006	2.000	1.992	0.002	2.000	1.994	0.000
2.5	Max	2.560	2.440	0.180	2.525	2.480	0.070	2.514	2.494	0.030	2.510	2.498	0.018	2.510	2.500	0.016
	Min	2.500	2.380	0.060	2.500	2.455	0.020	2.500	2.484	0.006	2.500	2.492	0.002	2.500	2.494	0.000
3	Max	3.060	2.940	0.180	3.025	2.980	0.070	3.014	2.994	0.030	3.010	2.998	0.018	3.010	3.000	0.016
	Min	3.000	2.880	0.060	3.000	2.955	0.020	3.000	2.984	0.006	3.000	2.992	0.002	3.000	2.994	0.000
4	Max	4.075	3.930	0.220	4.030	3.970	0.090	4.018	3.990	0.040	4.012	3.996	0.024	4.012	4.000	0.020
	Min	4.000	3.855	0.070	4.000	3.940	0.030	4.000	3.978	0.010	4.000	3.988	0.004	4.000	3.992	0.000
5	Max	5.075	4.930	0.220	5.030	4.970	0.090	5.018	4.990	0.040	5.012	4.996	0.024	5.012	5.000	0.020
	Min	5.000	4.855	0.070	5.000	4.940	0.030	5.000	4.978	0.010	5.000	4.988	0.004	5.000	4.992	0.000
6	Max	6.075	5.930	0.220	6.030	5.970	0.090	6.018	5.990	0.040	6.012	5.996	0.024	6.012	6.000	0.020
	Min	6.000	5.855	0.070	6.000	5.940	0.030	6.000	5.978	0.010	6.000	5.988	0.004	6.000	5.992	0.000
8	Max	8.090	7.920	0.260	8.036	7.960	0.112	8.022	7.987	0.050	8.015	7.995	0.029	8.015	8.000	0.024
	Min	8.000	7.830	0.080	8.000	7.924	0.040	8.000	7.972	0.013	8.000	7.986	0.005	8.000	7.991	0.000
10	Max	10.090	9.920	0.260	10.036	9.960	0.112	10.022	9.987	0.050	10.015	9.995	0.029	10.015	10.000	0.024
	Min	10.000	9.830	0.080	10.000	9.924	0.040	10.000	9.972	0.013	10.000	9.986	0.005	10.000	9.991	0.000
12	Max	12.110	11.905	0.315	12.043	11.950	0.136	12.027	11.984	0.061	12.018	11.994	0.035	12.018	12.000	0.029
	Min	12.000	11.795	0.095	12.000	11.907	0.050	12.000	11.966	0.016	12.000	11.983	0.006	12.000	11.989	0.000
16	Max	16.110	15.905	0.315	16.043	15.950	0.136	16.027	15.984	0.061	16.018	15.994	0.035	16.018	16.000	0.029

MATING DIMENSIONS:

CLASSES OF FITS (Clearance fits)

Dimensions are in millimeters.

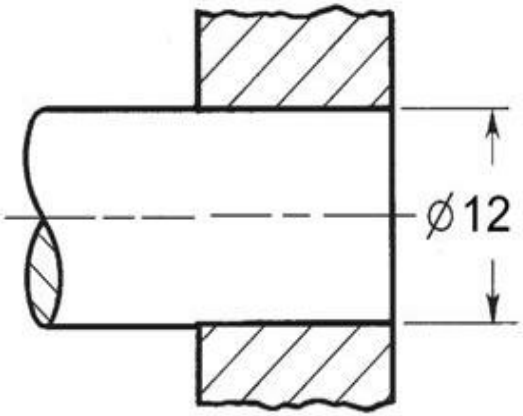
Basic Size		Loose Running			Free Running			Close Running			Sliding			Locational Clearance		
		Hole H11	Shaft c11	Fit	Hole H9	Shaft d9	Fit	Hole H8	Shaft f7	Fit	Hole H7	Shaft g6	Fit	Hole H7	Shaft h6	Fit
40	Max	40.160	39.880	0.440	40.062	39.920	0.204	40.039	39.975	0.089	40.025	39.991	0.050	40.025	40.000	0.041
	Min	40.000	39.720	0.120	40.000	39.858	0.080	40.000	39.950	0.025	40.000	39.975	0.009	40.000	39.984	0.000
50	Max	50.160	49.870	0.450	50.062	49.920	0.204	50.039	49.975	0.089	50.025	49.991	0.050	50.025	50.000	0.041
	Min	50.000	49.710	0.130	50.000	49.858	0.080	50.000	49.950	0.025	50.000	49.975	0.009	50.000	49.984	0.000
60	Max	60.190	59.860	0.520	60.074	59.900	0.248	60.046	59.970	0.106	60.030	59.990	0.059	60.030	60.000	0.049
	Min	60.000	59.670	0.140	60.000	59.826	0.100	60.000	59.940	0.030	60.000	59.971	0.010	60.000	59.981	0.000
80	Max	80.190	79.950	0.530	80.074	79.900	0.248	80.046	79.970	0.106	80.030	79.990	0.059	80.030	80.000	0.049
	Min	80.000	79.660	0.150	80.000	79.826	0.100	80.000	79.940	0.030	80.000	79.971	0.010	80.000	79.981	0.000
100	Max	100.220	99.830	0.610	100.087	99.880	0.294	100.054	99.964	0.125	100.035	99.988	0.069	100.035	100.000	0.057
	Min	100.000	99.610	0.170	100.000	99.793	0.120	100.000	99.929	0.036	100.000	99.966	0.012	100.000	99.978	0.000
120	Max	120.220	119.820	0.620	120.087	119.880	0.294	120.054	119.964	0.125	120.035	119.988	0.069	120.035	120.000	0.057
	Min	120.000	119.600	0.180	120.000	119.793	0.120	120.000	119.929	0.036	120.000	119.966	0.012	120.000	119.978	0.000
160	Max	160.250	159.790	0.710	160.100	159.855	0.345	160.063	159.957	0.146	160.000	159.986	0.079	160.040	160.000	0.065
	Min	160.000	159.540	0.210	160.000	159.755	0.145	160.000	159.917	0.043	160.000	159.961	0.014	160.000	159.975	0.000
200	Max	200.290	199.760	0.820	200.115	199.830	0.400	200.072	199.950	0.168	200.046	199.985	0.090	200.046	200.000	0.075
	Min	200.000	199.470	0.240	200.000	199.715	0.170	200.000	199.904	0.050	200.000	199.956	0.015	200.000	199.971	0.000
250	Max	250.290	249.720	0.860	250.115	249.830	0.400	250.072	249.950	0.168	250.046	249.985	0.090	250.046	250.000	0.075
	Min	250.000	249.430	0.280	250.000	249.715	0.170	250.000	249.904	0.050	250.000	249.956	0.015	250.000	249.971	0.000
300	Max	300.320	299.670	0.970	300.130	299.810	0.450	300.081	299.944	0.189	300.052	299.983	0.101	300.052	300.000	0.084
	Min	300.000	299.350	0.330	300.000	299.680	0.190	300.000	299.892	0.056	300.000	299.951	0.017	300.000	299.968	0.000
400	Max	400.360	399.600	1.120	400.140	399.790	0.490	400.089	399.938	0.208	400.057	399.982	0.111	400.057	400.000	0.093
	Min	400.000	399.240	0.400	400.000	399.650	0.210	400.000	399.881	0.062	400.000	399.946	0.018	400.000	399.964	0.000
500	Max	500.400	499.520	1.280	500.155	499.770	0.540	500.097	499.932	0.228	500.063	499.980	0.123	500.063	500.000	0.103
	Min	500.000	499.120	0.480	500.000	499.615	0.230	500.000	499.869	0.068	500.000	499.940	0.020	500.000	499.960	0.000

[a] From ANSI B4.2–1978 (R1994). For description of preferred fits, see Table 12.2.

CLASSES OF FITS (Clearance fits)

Example:

For free running clearance fit, find hole and shaft limits for a base size of 12mm. Also provide the fit symbol in the preferred form.



Solution:

From the previous table for free running fit,

Hole limits: 12.000-12.043

Shaft limits: 11.907-11.950

Fit symbol: 12H9/d9

Basic Size		Loose Running			Free Running			Close Running			Sliding			Locational Clearance		
		Hole H11	Shaft c11	Fit	Hole H9	Shaft d9	Fit	Hole H8	f7	Fit	Hole H7	Shaft g6	Fit	Hole H7	Shaft h6	Fit
10	Max	10.090	9.920	0.260	10.036	9.960	0.112	10.022	9.987	0.050	10.015	9.995	0.029	10.015	10.000	0.024
	Min	10.000	9.830	0.080	10.000	9.924	0.040	10.000	9.972	0.013	10.000	9.986	0.005	10.000	9.991	0.000
12	Max	12.110	11.905	0.315	12.043	11.950	0.136	12.027	11.984	0.061	12.018	11.994	0.035	12.018	12.000	0.029
	Min	12.000	11.795	0.095	12.000	11.907	0.050	12.000	11.966	0.016	12.000	11.983	0.006	12.000	11.989	0.000
16	Max	16.110	15.905	0.315	16.043	15.950	0.136	16.027	15.984	0.061	16.018	15.994	0.035	16.018	16.000	0.029
	Min	16.000	15.795	0.095	16.000	15.907	0.050	16.000	15.966	0.016	16.000	15.983	0.006	16.000	15.989	0.000

CLASSES OF FITS (ISO SYMBOLS)

Interference fits:

- Location interference: H7/p6
- Slight interference (press fit). Parts to be assembled/disassembled with accuracy without bore pressure. Ex: couplings, brass collars…Medium drive: H7/s6
- Ordinary steel parts, shrink fit on light sections, tightest fit for cast iron.
- Force fit: H7/u6: parts to be joined with high stresses, shrink fit on medium sections.

Transition fits:

- Compromise between clearance and interference fits.
- For application where accuracy of location is important (small clearance or interference is permissible). Ex., gears and keys on shafts.
- Location transition: H7/k6
- Location transition: H7/n6 (more accurate location)

CLASSES OF FITS (ISO SYMBOLS)

CLASSES OF FITS (Transition & Interference Fits)

Dimensions are in millimeters.

Basic Size		Locational Transn.			Locational Transn.			Locational Interf.			Medium Drive			Force		
		Hole H7	Shaft k6	Fit	Hole H7	Shaft n6	Fit	Hole H7	Shaft p6	Fit	Hole H7	Shaft s6	Fit	Hole H7	Shaft u6	Fit
1	Max	1.010	1.006	0.010	1.010	1.010	0.006	1.010	1.012	0.004	1.010	1.020	−0.004	1.010	1.024	−0.008
	Min	1.000	1.000	−0.006	1.000	1.004	−0.010	1.000	1.006	−0.012	1.000	1.014	−0.020	1.000	1.018	−0.024
1.2	Max	1.210	1.206	0.010	1.210	1.210	0.006	1.210	1.212	0.004	1.210	1.220	−0.004	1.210	1.224	−0.008
	Min	1.200	1.200	−0.006	1.200	1.204	−0.010	1.200	1.206	−0.012	1.200	1.214	−0.020	1.200	1.218	−0.024
1.6	Max	1.610	1.606	0.010	1.610	1.610	0.006	1.610	1.612	0.004	1.610	1.620	−0.004	1.610	1.624	−0.008
	Min	1.600	1.600	−0.006	1.600	1.604	−0.010	1.600	1.606	−0.012	1.600	1.614	−0.020	1.600	1.618	−0.024
2	Max	2.010	2.006	0.010	2.010	2.010	0.006	2.010	2.012	0.004	2.010	2.020	−0.004	2.010	2.024	−0.008
	Min	2.000	2.000	−0.006	2.000	2.004	−0.010	2.000	2.006	−0.012	2.000	2.014	−0.020	2.000	2.018	−0.024
2.5	Max	2.510	2.506	0.010	2.510	2.510	0.006	2.510	2.512	0.004	2.510	2.520	−0.004	2.510	2.524	−0.008
	Min	2.500	2.500	−0.006	2.500	2.504	−0.010	2.500	2.506	−0.012	2.500	2.514	−0.020	2.500	2.518	−0.024
3	Max	3.010	3.006	0.010	3.010	3.010	0.006	3.010	3.012	0.004	3.010	3.020	−0.004	3.010	3.024	−0.008
	Min	3.000	3.000	−0.006	3.000	3.004	−0.010	3.000	3.006	−0.012	3.000	3.014	−0.020	3.000	3.018	−0.024
4	Max	4.012	4.009	0.011	4.012	4.016	0.004	4.012	4.020	0.000	4.012	4.027	−0.007	4.012	4.031	−0.011
	Min	4.000	4.001	−0.009	4.000	4.008	−0.016	4.000	4.012	−0.020	4.000	4.019	−0.027	4.000	4.023	−0.031
5	Max	5.012	5.009	0.011	5.012	5.016	0.004	5.012	5.020	0.000	5.012	5.027	−0.007	5.012	5.031	−0.011
	Min	5.000	5.001	−0.009	5.000	5.008	−0.016	5.000	5.012	−0.020	5.000	5.019	−0.027	5.000	5.023	−0.031
6	Max	6.012	6.009	0.011	6.012	6.016	0.004	6.012	6.020	0.000	6.012	6.027	−0.007	6.012	6.031	−0.011
	Min	6.000	6.001	−0.009	6.000	6.008	−0.016	6.000	6.012	−0.020	6.000	6.019	−0.027	6.000	6.023	−0.031
8	Max	8.015	8.010	0.014	8.015	8.019	0.005	8.015	8.024	0.000	8.015	8.032	−0.008	8.015	8.037	−0.013
	Min	8.000	8.001	−0.010	8.000	8.010	−0.019	8.000	8.015	−0.024	8.000	8.023	−0.032	8.000	8.028	−0.037
10	Max	10.015	10.010	0.014	10.015	10.019	0.005	10.015	10.024	0.000	10.015	10.032	−0.008	10.015	10.037	−0.013
	Min	10.000	10.001	−0.010	10.000	10.010	−0.019	10.000	10.015	−0.024	10.000	10.023	−0.032	10.000	10.028	−0.037
12	Max	12.018	12.012	0.017	12.018	12.023	0.006	12.018	12.029	0.000	12.018	12.039	−0.010	12.018	12.044	−0.015
	Min	12.000	12.001	−0.012	12.000	12.012	−0.023	12.000	12.018	−0.029	12.000	12.028	−0.039	12.000	12.033	−0.044

CLASSES OF FITS (ISO SYMBOLS)

CLASSES OF FITS (Transition & Interference Fits)

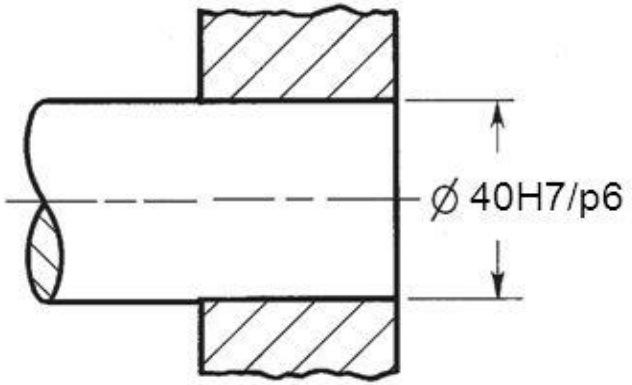
Dimensions are in millimeters.

Basic Size		Locational Transn.			Locational Transn.			Locational Interf.			Medium Drive			Force		
		Hole H7	Shaft k6	Fit	Hole H7	Shaft n6	Fit	Hole H7	Shaft p6	Fit	Hole H7	Shaft s6	Fit	Hole H7	Shaft u6	Fit
40	Max	40.025	40.018	0.023	40.025	40.033	0.08	40.025	40.042	−0.001	40.025	40.059	−0.018	40.025	40.076	−0.035
	Min	40.000	40.002	−0.018	40.000	40.017	−0.033	40.000	40.026	−0.042	40.000	40.043	−0.059	40.000	40.060	−0.076
50	Max	50.025	50.018	0.023	50.025	50.033	0.008	50.025	50.042	−0.001	50.025	50.059	−0.018	50.025	50.086	−0.045
	Min	50.000	50.002	−0.018	50.000	50.017	−0.033	50.000	50.026	−0.042	50.000	50.043	−0.059	50.000	50.070	−0.086
60	Max	60.030	60.021	0.028	60.030	60.039	0.010	60.030	60.051	−0.002	60.030	60.072	−0.023	60.030	60.106	−0.057
	Min	60.000	60.002	−0.021	60.000	60.020	−0.039	60.000	60.032	−0.051	60.000	60.053	−0.072	60.000	60.087	−0.106
80	Max	80.030	80.021	0.028	80.030	80.039	0.010	80.030	80.051	−0.002	80.030	80.078	−0.029	80.030	80.121	−0.072
	Min	80.000	80.002	−0.021	80.000	80.020	−0.039	80.000	80.032	−0.051	80.000	80.059	−0.078	80.000	80.102	−0.121
100	Max	100.035	100.025	0.032	100.035	100.045	0.012	100.035	100.059	−0.002	100.035	100.093	−0.036	100.035	100.146	−0.089
	Min	100.000	100.003	−0.025	100.000	100.023	−0.045	100.000	100.037	−0.059	100.000	100.071	−0.093	100.000	100.124	−0.146
120	Max	120.035	120.025	0.032	120.035	120.045	0.012	120.035	120.059	−0.002	120.035	120.101	−0.044	120.035	120.166	−0.109
	Min	120.000	120.003	−0.025	120.000	120.023	−0.045	120.000	120.037	−0.059	120.000	120.079	−0.101	120.000	120.144	−0.166
160	Max	160.040	160.028	0.037	160.040	160.052	0.013	160.040	160.068	−0.003	160.040	160.125	−0.060	160.040	160.215	−0.150
	Min	160.000	160.003	−0.028	160.000	160.027	−0.052	160.000	160.043	−0.068	160.000	160.100	−0.125	160.000	160.190	−0.215
200	Max	200.046	200.033	0.042	200.046	200.060	0.015	200.046	200.079	−0.004	200.046	200.151	−0.076	200.046	200.265	−0.190
	Min	200.000	200.004	−0.033	200.000	200.031	−0.060	200.000	200.050	−0.079	200.000	200.122	−0.151	200.000	200.236	−0.265
250	Max	250.046	250.033	0.042	250.046	250.060	0.015	250.046	250.079	−0.004	250.046	250.169	−0.094	250.046	250.313	−0.238
	Min	250.000	250.004	−0.033	250.000	250.031	−0.060	250.000	250.050	−0.079	250.000	250.140	−0.169	250.000	250.284	−0.313
300	Max	300.052	300.036	0.048	300.052	300.066	0.018	300.052	300.088	−0.004	300.052	300.202	−0.118	300.052	300.382	−0.298
	Min	300.000	300.004	−0.036	300.000	300.034	−0.066	300.000	300.056	−0.088	300.000	300.170	−0.202	300.000	300.350	−0.382
400	Max	400.057	400.040	0.053	400.057	400.073	0.020	400.057	400.098	−0.005	400.057	400.244	−0.151	400.057	400.471	−0.378
	Min	400.000	400.004	−0.040	400.000	400.037	−0.073	400.000	400.062	−0.098	400.000	400.208	−0.244	400.000	400.435	−0.471
500	Max	500.063	500.045	0.058	500.063	500.080	0.023	500.063	500.108	−0.005	500.063	500.292	−0.189	500.063	500.580	−0.477
	Min	500.000	500.005	−0.045	500.000	500.040	−0.080	500.000	500.068	−0.108	500.000	500.252	−0.292	500.000	500.540	−0.580

CLASSES OF FITS (Transition & Interference Fits)

Example:

For a basic hole diameter of 40mm, H7p6, determine the class of the fit, the limits for the hole and the shaft.



Solution:

From the previous table for free running fit,

Type of fit: locational interference

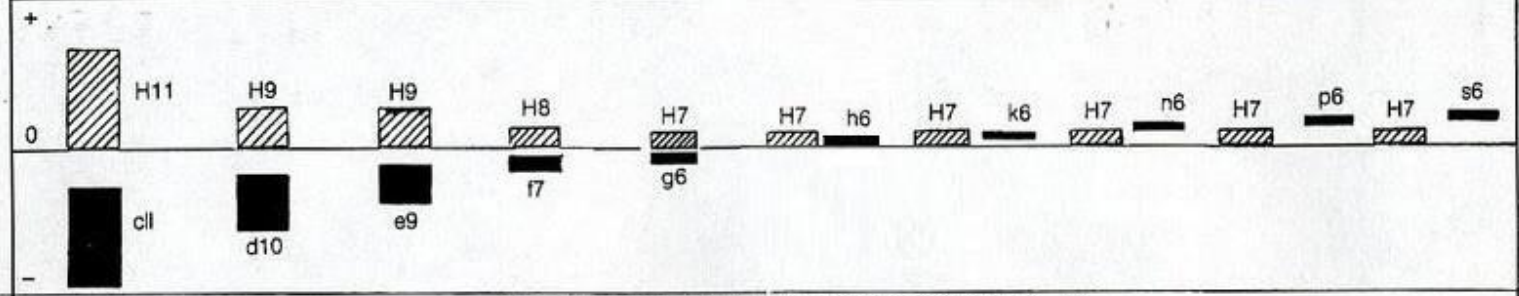
Hole limits: 40.000 – 40.025

Shaft limits: 40.026 - 40.042

Basic Size		Locational Transn.			Locational Transn.			Locational Interf.			Medium Drive			Force		
		Hole H7	Shaft k6	Fit	Hole H7	Shaft n6	Fit	Hole H7	Shaft p6	Fit	Hole H7	Shaft s6	Fit	Hole H7	Shaft u6	Fit
40	Max	40.025	40.018	0.023	40.025	40.033	0.08	40.025	40.042	−0.001	40.025	40.059	−0.018	40.025	40.076	−0.035
	Min	40.000	40.002	−0.018	40.000	40.017	−0.033	40.000	40.026	−0.042	40.000	40.043	−0.059	40.000	40.060	−0.076
50	Max	50.025	50.018	0.023	50.025	50.033	0.008	50.025	50.042	−0.001	50.025	50.059	−0.018	50.025	50.086	−0.045
	Min	50.000	50.002	−0.018	50.000	50.017	−0.033	50.000	50.026	−0.042	50.000	50.043	−0.059	50.000	50.070	−0.086
60	Max	60.030	60.021	0.028	60.030	60.039	0.010	60.030	60.051	−0.002	60.030	60.072	−0.023	60.030	60.106	−0.057
	Min	60.000	60.002	−0.021	60.000	60.020	−0.039	60.000	60.032	−0.051	60.000	60.053	−0.072	60.000	60.087	−0.106

TRANSITION FITS AND INTERFERENCE FITS

Combination of Hole and Shaft	Quality of Fit	Typical uses
H6 j5 fine	Clearance transition	Very small clearance is obtained - used for fits where a slight interference is permissible - coupling spigots and recesses, gear rings clamped to steel hubs.
H7 j6 normal		
H8 j7 coarse		
H6 k5 fine	True transition	Fits averaging no clearance where slight interference can be tolerated , with the object of eliminating vibration - ball bearings races of light duty.
H7 k6 normal		
H8 k7 coarse		
H6 m5 fine	Interference transition	Fits averaging slight interference - used for ball bearing races of medium duty.
H7 m6 normal		
H8 m7 coarse		
H6 p5 fine	Press fit	Light press fit for nonferrous parts which can be dismantled when required-bearing bushes-press fit for steel, cast iron or brass to steel assemblies- bush in a gear.
H7 p6 normal		
H6 r5 fine	Drive fit	Medium drive fit for ferrous parts and light drive fit for non-ferrous parts that can be dismantled.
H7 r6 normal		
H6 s5 fine	Drive fit	Permanent or semi permanent assemblies of steel and cast iron with considerable gripping force- collars pressed on to shafts, valve seatings etc. For light alloys this gives a press fit.
H7 s6 normal		
H6 u5 fine	Force or shrink fit	High interference fit a thorough investigation into the degree of grip and the stresses in the part must be made.
H7 u6 normal		



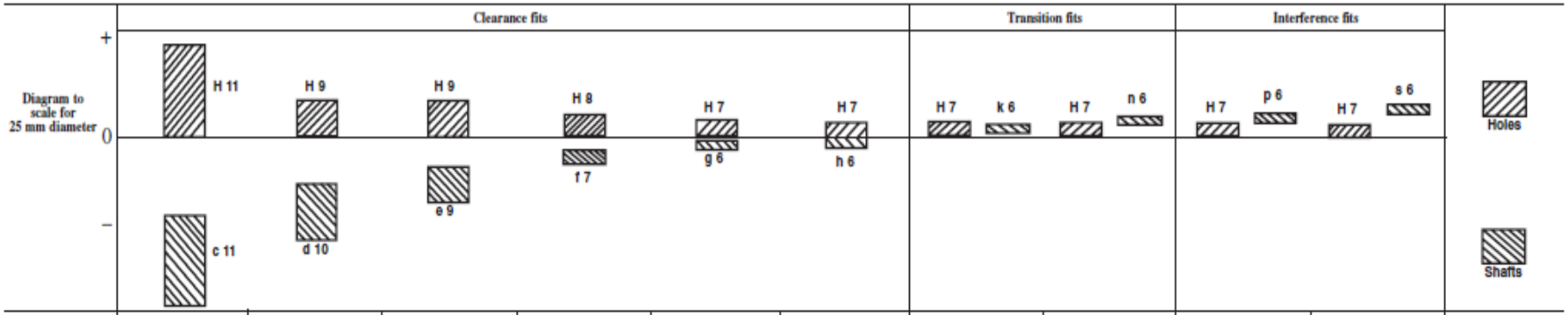
Basic size (mm)		Upper and lower deviations for tolerance class (Values μm)																				Basic size (mm)	
		H11	c11	H9	d10	H9	e9	H8	f7	H7	g6	H7	h6	H7	k6	H7	n6	H7	p6	H7	s6		
Above	Up to and incl.	+	–	+	–	+	–	+	–	+	–	+	+	+	+	+	+	+	+	+	+	Above	Up to and incl.
0	3	60 0	60 120	25 0	20 60	25 0	14 39	14 0	6 16	10 0	2 8	10 0	6 0	10 0	6 0	10 0	10 4	10 0	12 8	10 0	20 14	0	3
3	6	75 0	70 145	30 0	30 78	30 0	20 50	18 0	10 22	12 0	4 12	15 0	9 0	15 0	10 1	15 0	19 10	15 0	24 15	15 0	32 23	3	6
6	10	90 0	80 170	36 0	40 98	36 0	25 61	22 0	13 28	15 0	5 14	18 0	11 0	18 0	12 1	18 0	23 12	18 0	29 18	18 0	39 28	6	10
10	18	110 0	95 205	43 0	50 120	43 0	32 75	27 0	16 34	18 0	6 17	21 0	13 0	21 0	15 2	21 0	28 15	21 0	35 22	21 0	48 35	10	18
18	30	130 0	110 240																			18	30
30	40	160 0	120 280	62 0	80 180	62 0	50 112	39 0	25 50	25 0	9 25	25 0	16 0	25 0	18 2	25 0	33 17	25 0	42 26	25 0	59 43	30	40
40	50	160 0	130 290																			40	50
50	65	190 0	140 330	74 0	100 220	74 0	60 134	46 0	30 60	30 0	10 29	30 0	19 0	30 0	21 2	30 0	39 20	30 0	51 32	30 0	72 53	50	65
65	80	190 0	150 340																	30 0	78 59	65	80
80	100	220 0	170 390	87 0	120 260	87 0	72 159	54 0	36 71	35 0	12 34	35 0	22 0	35 0	25 3	35 0	45 23	35 0	59 37	35 0	93 71	80	100
100	120	220 0	180 400																	35 0	101 79	100	120
120	140	250 0	200 450	100 0	145 305	100 0	84 185	63 0	43 83	40 0	14 39	40 0	25 0	40 0	28 3	40 0	52 27	40 0	68 43	40 0	117 92	120	140
140	160	250 0	210 460																	40 0	125 100	140	160
160	180	250 0	230 480																	40 0	133 108	160	180
180	200	290 0	240 530	115 0	170 355	115 0	100 215	72 0	50 96	46 0	15 44	46 0	29 0	46 0	33 4	46 0	60 31	46 0	79 50	46 0	151 122	180	200
200	225	290 0	260 550																	46 0	159 130	200	225
225	250	290 0	280 570																	46 0	169 140	225	250
250	280	320 0	300 630	130 0	190 400	130 0	110 240	81 0	56 108	52 0	17 49	52 0	32 0	52 0	36 4	52 0	66 34	52 0	88 56	52 0	190 158	250	280
280	315	320 0	330 650																	52 0	202 170	280	315
315	355	360 0	360 720	140 0	210 440	140 0	125 265	89 0	62 119	57 0	18 54	57 0	36 0	57 0	40 4	57 0	73 37	57 0	98 62	57 0	226 190	315	355
355	400	360 0	400 760																	57 0	244 208	355	400
400	450	400 0	440 840	155 0	230 480	155 0	135 290	97 0	68 131	63 0	20 60	63 0	40 0	63 0	45 5	63 0	80 40	63 0	108 68	63 0	272 232	400	450
450	500	400 0	480 880																	63 0	292 252	450	500

Extracted from BS 4500 : 1969

BRITISH STANDARD

SELECTED ISO FITS—HOLE BASIS

Data Sheet 4500A
Issue 1. February 1970
confirmed August 1985



Nominal sizes		Tolerance		Tolerance		Tolerance		Tolerance		Tolerance		Tolerance		Tolerance		Tolerance		Tolerance		Tolerance		Tolerance		Nominal sizes	
Over	To	H11	c11	H9	d10	H9	e9	H8	f7	H7	g6	H7	h6	H7	k6	H7	n6	H7	p6	H7	s6	Over	To
mm	mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	0.001 mm	mm	mm
—	3	+60 0	−60 −120	+25 0	−20 −60	+25 0	−14 −39	+14 0	−6 −16	+10 0	−2 −8	+10 0	−6 0	+10 0	+6 +0	+10 0	+10 +4	+10 0	+12 +6	+10 0	+20 +14	—	3
3	6	+75 0	−70 −145	+30 0	−30 −78	+30 0	−20 −50	+18 0	−10 −28	+12 0	−4 −12	+12 0	−8 0	+12 0	+9 +1	+12 0	+16 +8	+12 0	+20 +12	+12 0	+27 +19	3	6
6	10	+90 0	−80 −170	+36 0	−40 −98	+36 0	−25 −61	+22 0	−13 −28	+15 0	−5 −14	+15 0	−9 0	+15 0	+10 +1	+15 0	+19 +10	+15 0	+24 +15	+15 0	+32 +23	6	10
10	18	+110 0	−95 −205	+43 0	−50 −120	+43 0	−32 −75	+27 0	−16 −34	+18 0	−6 −17	+18 0	−11 0	+18 0	+12 +1	+18 0	+23 +12	+18 0	+29 +18	+18 0	+39 +28	10	18
18	30	+130 0	−110 −240	+52 0	−65 −149	+52 0	−40 −92	+33 0	−20 −41	+21 0	−7 −20	+21 0	−13 0	+21 0	+15 +2	+21 0	+28 +15	+21 0	+35 +22	+21 0	+48 +35	18	30
30	40	+160 0	−120 −280	+62	−80	+62	−50	+39	−25	+25	−9	+25	−16	+25	+18	+25	+33	+25	+42	+25	+59	30	40
40	50	+160 0	−130 −290	0	−180	0	−112	0	−50	0	−25	0	0	0	+2	0	+17	0	+26	0	+43	40	50
50	65	+190 0	−140 −330	+74	−100	−74	−60	+46	−30	+30	−10	+30	−19	+30	+21	+30	+39	+30	+51	+30 0	+72 +53	50	65
65	80	+190 0	−150 −340	0	−220	0	−134	0	−60	0	−29	0	0	0	+2	0	+20	0	+32	+30 0	+78 +59	65	80
80	100	+220 0	−170 −390	+87	−120	−87	−72	+54	−36	+35	−12	+35	−22	+35	+25	+35	+45	+35	+59	+35 0	+93 +71	80	100
100	120	+220 0	−180 −400	+0	−260	−0	−159	0	−71	0	−34	0	0	0	+3	0	+23	0	+37	+35 +0	+101 +79	100	120
120	140	+250 0	−200 −450																	+40 0	+117 +92	120	140
140	160	+250 0	−210 −460	+100 0	145 305	+100 0	−84 −185	+63 0	−43 −83	−40 0	−14 −39	+40 0	−25 0	+40 0	+28 +3	+40 0	+52 +27	−40 0	+68 +43	+40 0	+125 +100	140	160
160	180	+250 0	−230 −480																	+40 0	+133 +108	160	180
180	200	+290 0	−240 530																	+46 0	+151 +122	180	200
200	225	+290 0	260 550	+115 0	−170 −355	+115 0	−100 −215	+72 0	−50 −96	+46 0	−15 −44	+46 0	−29 0	+46 0	+33 +4	+46 0	+60 +31	+46 0	+79 +50	+46 0	+159 +130	200	225
225	250	+290 0	280 570																	+46 0	+169 +140	225	250
250	280	+320 0	300 620	+130	−190	+130	−110	+81	−56	+52	−17	+52	+32	+52	−36	+52	+66	+52	+88	+52 0	+190 +158	250	280
280	315	+320 0	330 650	0	−400	0	−240	0	−108	0	−49	0	0	0	+4	0	+34	0	+56	+52 0	+202 +170	280	315
315	355	+360 0	360 −720	+140	−210	+140	−125	+89	−62	+57	−18	+57	−36	+57	+40	+57	+73	+57	+98	+57 0	+226 +190	315	355
355	400	+360 0	400 760	0	−440	0	−265	0	−119	0	−54	0	0	0	+4	0	+37	0	+62	+57 0	+244 +208	355	400
400	450	+400 0	440 840	+155	−230	+155	−135	+97	−68	+63	−20	+63	−40	+63	+45	+63	+80	+63	+108	+63 0	+272 +232	400	450
450	500	+400 0	480 880	0	−480	0	−290	0	−131	0	−60	0	0	0	+5	0	+40	0	+68	+63 0	+292 +252	450	500